

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and computational statistician specializing in scalable likelihood inference, predictive modeling, and statistical diagnostics for dependent data. Forms testable hypotheses, designs rigorous simulation and empirical tests, and builds reproducible Python/PyTorch research systems for large real-world datasets, with C++/pybind11 integration for performance-critical computation.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, R, SQL, Linux, Git, AWS EC2, HPC/SLURM; data pipelines, research-code debugging, CPU/GPU computing, and backtesting

Quantitative Methods: Hypothesis testing, predictive modeling, regression and machine learning, probability and stochastic processes, Monte Carlo simulation, likelihood inference, MLE/GLS, numerical optimization, time-series and spatiotemporal modeling, spectral and autocorrelation analysis, model validation, and performance evaluation

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Translated a sequential risk-monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and measurable decision rules for binary indicators.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, uncertainty, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2; presented findings as pricing and risk-segmentation recommendations to business stakeholders.

INDEPENDENT QUANTITATIVE RESEARCH

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Statistical Diagnostics

Sep 2024 - Present

- Scalable Inference:** Developed an advection-aware Vecchia likelihood approximation for nonseparable space-time covariance structures, using ordered local conditional models for likelihood-based parameter estimation and uncertainty quantification with up to 145,008 observations per day.
- Stochastic Diagnostics:** Developed statistical diagnostics for dependent stochastic processes that localize where a fitted model fails across scale and frequency, distinguishing persistent dependence from high-frequency noise, missingness, acquisition artifacts, and covariance misspecification.
- Data and Research Systems:** Built reusable Python/PyTorch pipelines for quality filtering, missingness, irregular-grid matching, time-dependent alignment, optimization, simulation, and restartable CPU/GPU and HPC execution; integrated performance-critical C++ ordering routines through pybind11.

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and backtested an inventory-aware market-making strategy around empirically calibrated reference values; tracked position-limit saturation over a rolling 10-step window and triggered more aggressive inventory-reducing quotes to manage persistent exposure.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)